

DRY-RUN GUIDE | distance.java
Source: java\1st_belt\distance.java

CORE BEHAVIOR

calculateDistance applies the Euclidean distance formula; mainFunction reads four coordinates and prints the result to two decimals.

REPRESENTATIVE INPUT / CALL

Coordinates (0,0) and (3,4).

STEP TRACE

dx=3, dy=4; $\text{sqrt}(3^2+4^2)=\text{sqrt}(25)=5.0$; printf formats two places.

EXPECTED OUTPUT / RETURN

'5.00'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Challenge #3
003 |
004 | Write a program to calculate the distance between two points in a 2D plane.
005 |
006 | The distance between two points (x1, y1) and (x2, y2) in a plane is given by the
    | formula:
007 |  $D = \text{sqrt}((x2 - x1)^2 + (y2 - y1)^2)$ 
008 |
009 | Where:
010 | - (x1, y1) and (x2, y2) are the coordinates of the two points.
011 | - D is the distance between the two points.
012 |
013 | Input Format:
014 | The program will take four integer inputs: x1, y1, x2, y2, each on a separate line,
015 | representing the coordinates of two points.
016 |
017 | Output Format:
018 | The program will output the distance between the two points, rounded to two decimal
    | places.
019 | */
020 |
021 | import java.util.Scanner;
022 |
023 | public class Solution {
024 |     public static double calculateDistance(double x1, double y1, double x2, double
    | y2) {
025 |         // Implement the formula here
026 |         double distance = Math.sqrt(Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2));
027 |         return distance;
028 |     }
029 |
030 |     public static void mainFunction() {
031 |         Scanner sc = new Scanner(System.in);
032 |
033 |         double x1 = sc.nextDouble();
034 |         double y1 = sc.nextDouble();
```

```
035 |         double x2 = sc.nextDouble();
036 |         double y2 = sc.nextDouble();
037 |
038 |         double distance = calculateDistance(x1, y1, x2, y2);
039 |         System.out.printf("%.2f\n", distance);
040 |
041 |         sc.close();
042 |     }
043 |
044 |     public static void main(String[] args) {
045 |         mainFunction();
046 |     }
047 | }
048 |
```